

St. Francis Institute of Technology

Department of Information Technology

Mini Project – 2B Web Based on ML (ITM 601)

EVBHARAT

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Introduction

EVBHARAT

The worldwide electric vehicle (EV) market is growing rapidly (e.g. 14 million units sold in 2023), and India has also seen strong EV sales growth (rising from ~50,000 in 2016 to 2.08 million in 2024). However, EV adoption faces critical challenges. Limited driving range, long charging times and inadequate charging infrastructure remain key barriers. In particular, maintaining lithium-ion battery health (state-of-health, SoH) is essential for EV performance and safety. At the same time, drivers experience “range anxiety” when route planning ignores battery limits and charger availability. EVBharat is proposed as a web-based ML system that simultaneously monitors battery health and provides intelligent route optimization for EVs in India, addressing these gaps in a unified platform.

Need of the project

- **EV Safety & Accident Response**: Road accidents involving EVs require rapid detection and response. The absence of intelligent accident alert systems leads to delayed emergency assistance and increased risk to passengers and nearby road users.
- **Battery Health & Range Anxiety**: EV users lack real-time visibility into battery State of Health (SoH), resulting in unexpected failures, reduced range, and high maintenance costs.
- **Inefficient & Unsafe Routing**: Existing navigation systems ignore EV-specific constraints such as battery level, traffic conditions, and safety risks, often causing energy loss and delays.
- **Object Detection for Safety**: Lack of real-time object and pedestrian awareness increases accident risk, especially in dense Indian traffic conditions.
- **Indian Context Requirement**: India's congested roads, mixed traffic, and limited charging infrastructure demand an AI-based, India-specific solution that integrates accident alerts, object detection, battery health monitoring, and intelligent routing into a single platform.

Problem Statement

Electric vehicle (EV) users currently lack an integrated, intelligent software system that can monitor battery health, generate timely alerts, and plan energy-efficient routes. Existing navigation tools focus mainly on distance or time and ignore battery degradation, traffic conditions, safety risks, and energy consumption, leading to unexpected battery drain and reduced battery life. Battery State of Health (SoH) degradation and critical conditions often go unnoticed until failures occur, and there is limited support for intelligent alerts or safety awareness. Moreover, no AI-based, web-enabled solution tailored to Indian EV conditions combines battery health prediction, intelligent routing, and real-time alerts in a single platform. EV Bharat addresses this gap by using machine learning to predict battery health, detect critical conditions, generate intelligent alerts, and recommend optimal routes under battery, traffic, and safety constraints.

Objectives

Battery SoH Prediction: Develop an AI/ML model to estimate Li-ion battery state-of-health in real time, using data from battery management systems (voltage, current, temperature, cycles).

Route Optimization: Design an intelligent routing algorithm that incorporates battery state, traffic, elevation, and charging station availability to minimize travel time and energy use.

Web-Based Platform: Implement a user-friendly web interface where EV owners or fleet managers can input telemetry and destination, and receive battery health reports and optimized routes.

Integration and Alerts: Provide predictive alerts (e.g. “battery health below threshold”, “recommended charging stop”) and visualize energy usage along routes.

Performance Evaluation: Validate the system with benchmark metrics (SoH prediction accuracy, route efficiency) and compare against baseline (no optimization).

Method/Approach

Prediction Accuracy: Battery SoH model should achieve high accuracy (e.g. RMSE <5% SoH) and explainability. Literature reports SoH model ; our goal is comparable accuracy on test data.

Routing Efficiency: Target ≥15–20% improvement in travel time and energy over naive routing (based on benchmarks). The system should reduce charging wait times significantly (e.g. >50%).

User Benefits: EVBharat can extend battery life, reduce range anxiety, and improve trip reliability. As an example, advanced EV routing systems cut "battery violations" from ~18% to ~2%.

Robustness: The web app must respond in real time (<5 s per query) and handle typical workloads. It should degrade gracefully with missing data .

Scalability: Designed to scale for multiple users/vehicles. Extensible to new data sources (e.g. new charger locations, updated ML models).

Resources

Software

- Machine Learning Libraries: TensorFlow / PyTorch, FastAPI, Scikit-learn for classical ML.
- Machine Learning Algorithm:
  -**Random Forest:**Continuously monitors battery SoC & SoH and predicts remaining range using Machine Learning.
  -**Object Detection for Safety**:Uses computer vision models such as OpenCV / YOLO to detect pedestrians, vehicles, and obstacles, enabling safer navigation and alert generation.
  -**Intelligent Route Optimization**:Applies *Shortest Path Algorithms* with traffic data to select optimal routes.
  -**Low Battery Condition Handling**:When battery falls below a threshold:Triggers specific low-battery alerts ,automatically switches to energy-efficient routing mode and Recommends nearest charging options

Resources

-**Smart Alert & Decision System**:Generates real-time alerts, intelligent routing decisions, and user recommendations through a unified software platform based on battery status, traffic conditions, and system predictions.

- Database Management:Uses SQLite for development and testing, and MySQL for deployment-level data storage, including battery records, route information, alerts, and user data.
- Web Technologies:For frontend used ReactJS, Tailwind CSS, JavaScript and backend used Node.js which enables a responsive, scalable, and user-friendly web application.
- Mapping & Traffic APIs:Integrates Google Maps API / OSRM / TomTom API for route planning, traffic-aware navigation, and map visualization.

Risks and Dependencies

- **Data Availability:** Real battery data may be limited.
  Mitigation: Use public/synthetic datasets with transfer learning; incrementally add live data.
- **Model Accuracy:** Predictions may not generalize to new battery chemistries.
  Mitigation: Begin with generic Li-ion models and calibrate using local samples.
- **Integration Complexity:** Linking IoT, web app, and APIs is complex.
  Mitigation: Adopt a modular design and test components independently.
- **API Dependencies:** Third-party map/traffic APIs may face outages or rate limits.
  Mitigation: Use multiple providers and implement caching/fallbacks.
- **EV Adoption Rate:** Market changes may affect relevance.
  Remark: Range optimization and battery health remain broadly valuable.
- **Regulatory & Privacy:** User data may raise privacy concerns.
  Mitigation: Anonymize data and apply encryption for sensitive information.

References

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